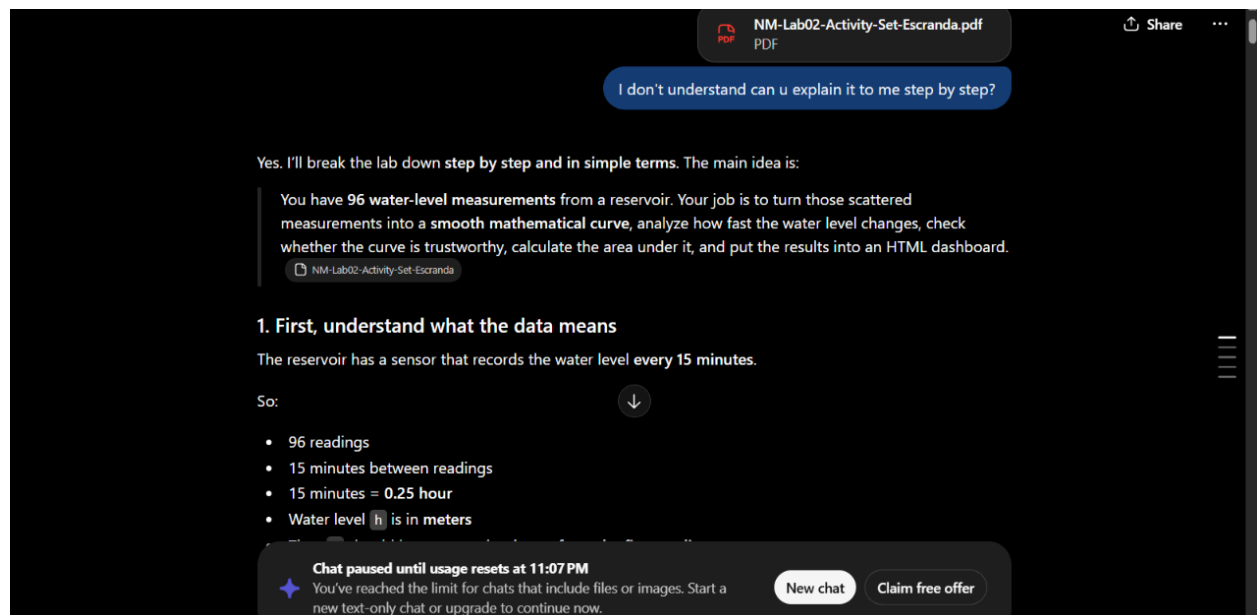


I USED CHAT GPT TO EXPLAIN TO ME THE STEPS TO MAKE IT CLEARER AND EASY TO UNDERSTAND, AND THEN I MAKE TO CHAT GPT TO GENERATE THE CODE I WILL USE TO THIS ACTIVITY.



Worked for 11s >

You're right — I should have given you the code directly.

I created the complete Python script using your 96 reservoir readings and the requirements from the laboratory PDF.

Python script

Download

lab02_Villasan.py

The script includes:

- All 96 reservoir readings
- Time conversion to hours
- Forward, backward, and central finite differences
- Second derivative
- 4-parameter logistic model
- Levenberg-Marquardt fitting with

Library / lab02_Villasan.py

```
"""
NM Laboratory Activity 02
Reservoir stage curve fitting using Levenberg-Marquardt.

Requirements:
    pip install numpy scipy

This script:
1. Stores the 96 reservoir readings.
2. Converts the 15-minute readings to elapsed hours.
3. Computes forward, backward, central, and second finite differences.
4. Fits a four-parameter logistic curve with scipy.optimize.curve_fit(method="lm").
5. Computes residuals, SSE, SST, R^2, standard error, parameter SE/t/p.
6. Integrates the fitted curve with scipy.integrate.quad.
7. Cross-checks the area with np.trapezoid.
8. Generates a self-contained HTML dashboard.

No fitting, regression, statistics, or numerical calculations are performed
by JavaScript. JavaScript is used only for switching dashboard tabs.
"""

import json
import html
from pathlib import Path

import numpy as np
from scipy.optimize import curve_fit
```

Python

Upgrade

Share

How to get the HTML using the Python code

Put these two files in the same folder.

lab02_Villasan.py

Then open Command Prompt in that folder and run:

Bash

python lab02_Villasan.py

After it finishes, the same folder will contain:

Your folder

lab02_Villasan.py

lab02_Villasan.html + THIS IS THE HTML

lab02_Villasan_results.txt

If you want the HTML be created in a specific folder

Ask anything

Think

Library / lab02_Villasan.py

```
"""
NM Laboratory Activity 02
Reservoir stage curve fitting using Levenberg-Marquardt.

Requirements:
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This script:
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by JavaScript. JavaScript is used only for switching dashboard tabs.
"""

import json
import html
from pathlib import Path

import numpy as np
from scipy.optimize import curve_fit
from scipy import stats
```